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A Growing Visible Light Leak in the 94 Å Channel of the Atmospheric Imaging Assembly

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ABSTRACT

The 94 Å channel of the Atmospheric Imaging Assembly (AIA) on the Solar Dynamics Observatory (SDO) is the instrument's main view of coronal plasma hotter than a few megakelvin, and with the other extreme ultraviolet (EUV) channels it underlies most differential emission measure (DEM) analyses of the corona. We find that the channel also records visible light from the photosphere. The leak is a sharp, limb-darkened image of the solar disk with a broad halo, and since each visible photon frees a single electron, it adds almost none of the photon noise that an EUV signal of the same size would. It is absent from the 304 Å channel, which shares the telescope and the detector, the thicker of the two focal-plane filters stops it, as its extra zirconium would stop visible light, and it can be seen directly as SDO passes into the shadow of the Earth, where it sets 76 s after the EUV has been absorbed, flattened by refraction like a sunset. In 28 such eclipses between 2010 and 2026, the leak at disk center grew from 0.030 DN s^{-1} to 0.90 DN s^{-1}, exponentially since 2016 with a doubling time of 3.1 yr, and it is now 66% of the quiet-Sun signal in the channel. Since 2019 it has been a constant part of the visible light that passes the entrance filters, which the instrument monitors with images taken without a focal-plane filter, and the noise of the channel itself gives the same history. Left in the data, the leak appears in DEMs as hot plasma in the quiet Sun and in coronal holes, it biases maps of Fe XVIII, and it hides part of the degradation of the channel from its calibration. We give a model of the leak and its history from which it can be subtracted.

Keywords: Solar instruments (1499), Calibration (2179), Solar extreme ultraviolet emission (1493), Solar corona (1483), Quiet sun (1322), Solar coronal holes (1484), Eclipses (442), Limb darkening (922)

1. INTRODUCTION

Outline.

- **The channel.** The Atmospheric Imaging Assembly (AIA) (J. R. Lemen et al. 2012) images the corona in seven extreme ultraviolet (EUV) channels from the Solar Dynamics Observatory (SDO) (W. D. Pesnell et al. 2012). The 94 Å channel sees Fe XVIII from plasma near 7 MK, and cooler lines of Fe IX and Fe X which atomic databases underestimate (P. Testa et al. 2012), so its temperature response was corrected empirically against the EUV Variability Experiment (EVE) (T. N. Woods et al. 2012; P. F. Boerner et al. 2014).

- **What depends on it.** It is the main constraint on hot plasma in differential emission measure (DEM) inversions (I. G. Hannah & E. P. Kontar 2012; J. Plowman et al. 2013; M. C. M. Cheung et al. 2015; H. Morgan & J. Pickering 2019; J. Plowman & A. Caspi 2020), and maps of Fe XVIII are made by subtracting the cool part of it predicted from the other channels (H. P. Warren et al. 2012; G. Del Zanna 2013).

- **Calibration over the mission.** The degradation of each channel is tracked against EVE and its successors (P. Boerner et al. 2012; P. F. Boerner et al. 2014; L. F. G. Dos Santos et al. 2021) and distributed in `aiapy` (W. Barnes et al. 2020). A signal that grows with time reads as a channel which degrades more slowly than it does.

- **Zirconium and visible light.** The 94 Å channel shares a telescope and a charge-coupled device (CCD) with 304 Å. Each has half of the aperture, behind its own entrance filter, and a filter wheel in front of the detector chooses between them with a focal-plane filter: zirconium for 94 Å and aluminum for 304 Å (J. R. Lemen et al. 2012). Zirconium has always passed some visible light. The zirconium entrance filters of the 94 and 131 Å channels pass enough of it to image the photosphere, and AIA monitors them routinely with images taken without a focal-plane filter (P. Boerner et al. 2012). It is the focal-plane filter that keeps this light from the detector, and the filter wheel holds

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a thicker spare in case the thin filter develops pinholes (J. R. Lemen et al. 2012).

- **Symptoms already in the literature.** In coronal holes the 94 Å signal matches what remains behind an occulter, and was taken for stray light (J. Saqri et al. 2020; S. G. Heinemann et al. 2021). A survey of quiet-Sun DEMs ended in 2018 because the reconstructions after it were poor (S. Purkhart & A. M. Veronig 2022), and some inversions drop the channel (S. J. Hofmeister et al. 2025). Coronal holes lose their contrast in it (S. M. Lezzi et al. 2023), and it is brighter than a DEM constrained in X-rays allows (P. S. Athiray & A. R. Winebarger 2024).
- **Roadmap.** Section 2 finds a signal in the channel that carries no photon noise, Section 3 sees it alone in eclipses and measures its history over the mission, Section 4 follows its path through the instrument with the two focal-plane filters, the images taken without one, and the other channels, Section 5 follows it into DEMs, maps of Fe XVIII, and the calibration of the channel, Section 6 gives a correction, and Section 7 concludes.

2. PHOTON NOISE

Outline.

- **Data.** A run of 15 consecutive level 1 frames at noon on each day of an eclipse, returned to the units of the detector by multiplying each by the flat field it was divided by.
- **Method.** The residual of each frame from the mean of its two neighbors cancels the Sun and anything that changes linearly with time, and has 3/2 the variance of one frame. It is sorted by the signal of the other frames of the run, whose noise is independent of it. The read noise, about 1.2 DN^2, comes from the corners of the CCD beyond the field stop; each quadrant of the CCD is read out by its own amplifier, and is fit on its own.
- **The noiseless signal.** Off the limb the variance rises from the read noise with the signal, as photon noise alone would, at the slope found near disk center. Near disk center the line of slope 0.54 DN meets the read noise only at a signal of 1.9 DN$\pm$0.2 DN: that much of the signal carries no photon noise (Figure 1, left). The photon noise of a signal grows with the electrons each photon frees in the silicon: dozens for a 94 Å photon, one for a visible photon, so a signal of visible light carries almost none.
- **Its map.** The noiseless signal fills a disk the size of the photosphere (Figure 1, middle), and does not follow the corona: across the disk it grows by 0.02 per DN of the 94 Å signal of a region, though it is 63% of that signal. It darkens toward the limb as $1 - u(1 - \mu)$ with $u = 0.57$ (Figure 1, right), where $\mu = \sqrt{1 - r^2}$ is the cosine of the angle between the line of sight and the vertical. This is limb darkening, the signature of an optically thick photosphere: we see into it only to a fixed optical depth along the line of sight, which near the limb, at a grazing angle, lies in higher and cooler layers, so the disk dims toward its edge. The optically thin corona does the opposite, brightening toward the limb, where the line of sight through it is longest. The channel sees the photosphere, in light that frees few electrons per photon: a visible light leak. Cite a measured limb darkening of the photosphere in visible light (e.g., Neckel and Labs 1994) to compare u with.

3. EARTH ECLIPSES

Outline.

- **Eclipse seasons.** SDO passes behind the Earth once a day for a few weeks twice a year, and the seasons drift earlier from year to year.
- **Ingress.** The atmosphere absorbs the EUV high above the surface, while the visible disk sets only near it, so for a while each part of the disk is seen in visible light alone. The 171 Å channel marks where the EUV has gone (Figure 2).
- **Method.** Each bin of the disk on a clock started by its own 171 Å edge; a forward model of a limb-darkened disk setting behind the atmosphere, seen through a point spread function (PSF) with a sharp core and broad wings, plus whatever EUV is left, fit to the frames in which the EUV is gone. The leak reaches the same detector as the EUV, so the 94 Å frames are flat fielded as the EUV is, and multiplied by the flat field at disk center to keep the units of the detector there; in the units of the detector the leak carries the pattern of the flat field, with a step at the edge of each quadrant from the gain of its amplifier. The dark is the median of the umbra just after the whole disk has set, rather than minutes later, since the offset of each quadrant drifts.
- **Residuals.** On the plateau, the frames are brighter than the fit toward the right of the detector in every eclipse since 2016, whichever way the edge sweeps the disk, by 5.3% of the leak at disk center per solar radius in the latest (Figure 3); the earliest also show the offsets of the quadrants. Just after the EUV is gone the frames are brighter than the fit, most near solar maximum, as if some EUV of 94 Å were still setting; each bin samples it once a frame, at its own phase, which stripes the map across the sweep. Around the setting, the frames fall below the fit just before it and rise above it just after: the disk sets more softly than the error function of the model. Off the limb they rise briefly as the disk sets there, with the light of the photosphere refracted along the limb of the Earth, which the model leaves out. Find where the gradient comes from: the flat field, which is made for the EUV, or the path of the leak through the instrument.
- **Results.** The leak at disk center in 28 eclipses from 2010 to 2026: from 0.030 $\mathrm{DN\,s^{-1}}$ to 0.90 $\mathrm{DN\,s^{-1}}$, with a doubling time of 3.1 yr since 2016 (Figure 4). Its limb darkening, the sharpness of its limb, and the halo beyond it.

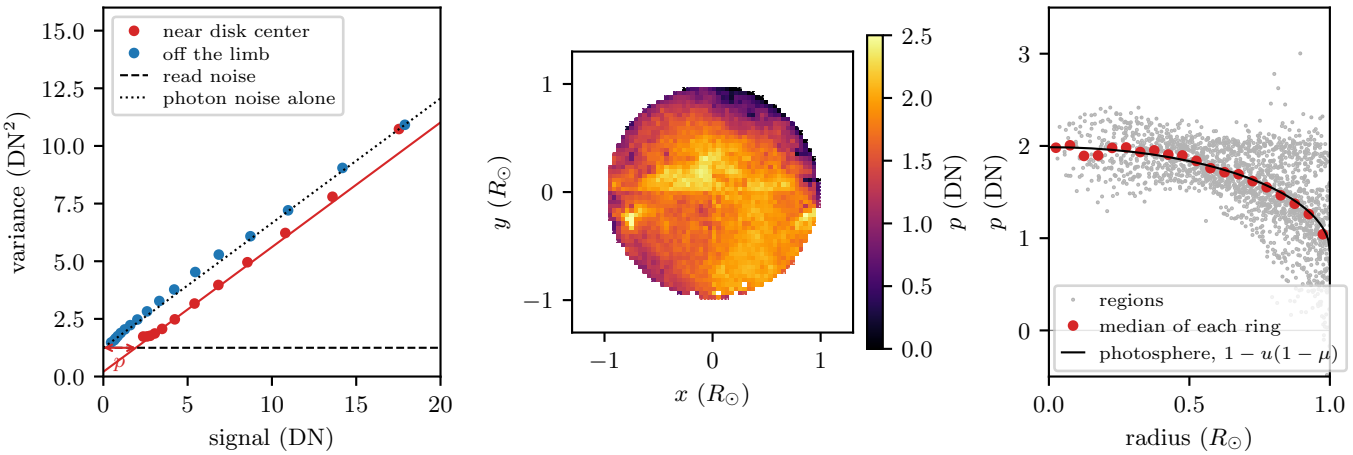


Figure 1. The noise of 94 Å in a run of 15 frames at noon on 2026 July 15, in the units of the detector. Left: the variance of one frame against its signal, near disk center (within half a radius) and off the limb (1.05 to 1.3 radii), and the read noise from the corners beyond the field stop. Off the limb the variance rises from the read noise; near disk center it reaches the read noise only at a signal of $p = 1.9$ DN $\pm$ 0.2 DN, which carries no photon noise. The dotted line is the variance if all of the signal carried photon noise, the slope near disk center started from the read noise at zero signal: the points off the limb follow it, and those near disk center lie p to its right. Each quadrant of the detector, read out by its own amplifier, is fit with its own read noise and slope; the line and the read noise shown are their means, and the uncertainty of p is its spread among the quadrants. Middle: p in each region of 64 × 64 pixels on the disk, with the read noise and the slope of its quadrant; off the limb the slope of the disk does not hold. Right: its radial profile, each region (gray) and the median of each ring of 0.05 $R_\odot$ (red), with the limb darkening of a photosphere, $1 - u(1 - \mu)$, where $\mu = \sqrt{1 - r^2}$ is the cosine of the angle between the line of sight and the vertical, fit to the regions within 0.95 $R_\odot$: $u = 0.57$.

- **Against the noise.** The noiseless signal near disk center in the run at noon of each day follows the leak (Figure 4). Since 2019, when the leak first exceeded three times the spread of the noise method among the quadrants (0.1 DN), the noiseless signal is 71% to 100% of the leak, the fraction falling as the leak grows; before, the leak is too faint for the noise to measure. Part of the shortfall is expected: a visible photon frees one electron, so the leak does carry a little photon noise, a twentieth of what an EUV signal of the same size would. Account for the rest: the rise of the variance per DN with signal, the step in the noiseless signal between the quadrants, and the change of the read noise in 2018.
- **Caveats.** The PSF is fit once to the brightest eclipses and held fixed, since in a single eclipse its core and wing trade off against each other, though the light they deliver to the disk does not. Before 2016 the leak is too faint to time its setting, which is held at its median since, and it lies above the exponential the later eclipses follow. One ingress fell during a flare.
- **Figures.** Still to come: the radial profile of the leak, and the sharpness of its limb.

4. INDEPENDENT CHECKS

Outline.

- **The focal-plane filters.** The channel has two focal-plane filters: the thin one it observes through, and a spare with 1000 Å more zirconium, in case the thin one develops pinholes (J. R. Lemen et al. 2012; P. Boerner et al. 2012). In the routine checks of the spare, each frame through it is followed by a frame of 304 Å through that channel's thick filter, and each carries a band of extra signal across the middle rows of the detector (Figure 5, left). The band is added as the frame is read out, while the filter wheel moves between the two thick filters: a frame through the spare followed by one through the thin filter of 304 Å has none (Figure 5, middle). On 2026 March 7, the day after a reset of the instrument, the channel observed through the spare all day, and its last frame through it was followed by one through the thin filter of 304 Å. In the frame before, the band reaches 1.7 DN, about as much as the leak through the thin filter at disk center, 1.9 DN. The last frame through the spare and the first 15 frames through the thin filter, which began 24 s later, are compared within 0.9 $R_\odot$ of disk center. There the spare passes 0.67 of the EUV, evenly across the field, measured from the structure of the corona, where the calibration of the channel expects 0.72. Of the leak, measured from the limb-darkened disk that the frame through the spare lacks, it passes -0.10 ± 0.05, the spread among the quadrants of the detector (Figure 5, right): none of it. The leak through the spare comes out negative; the noiseless signal through the thin filter may read low, as it does against the eclipses (Section 3). Light that crossed the metal of the filter would be attenuated by the extra zirconium by $e^{-4\pi k\,\Delta d/\lambda}$, where k is the extinction coefficient of zirconium at the wavelength λ and Δd is the extra thickness. For the EUV this is 0.72 with the optical constants of B. L.

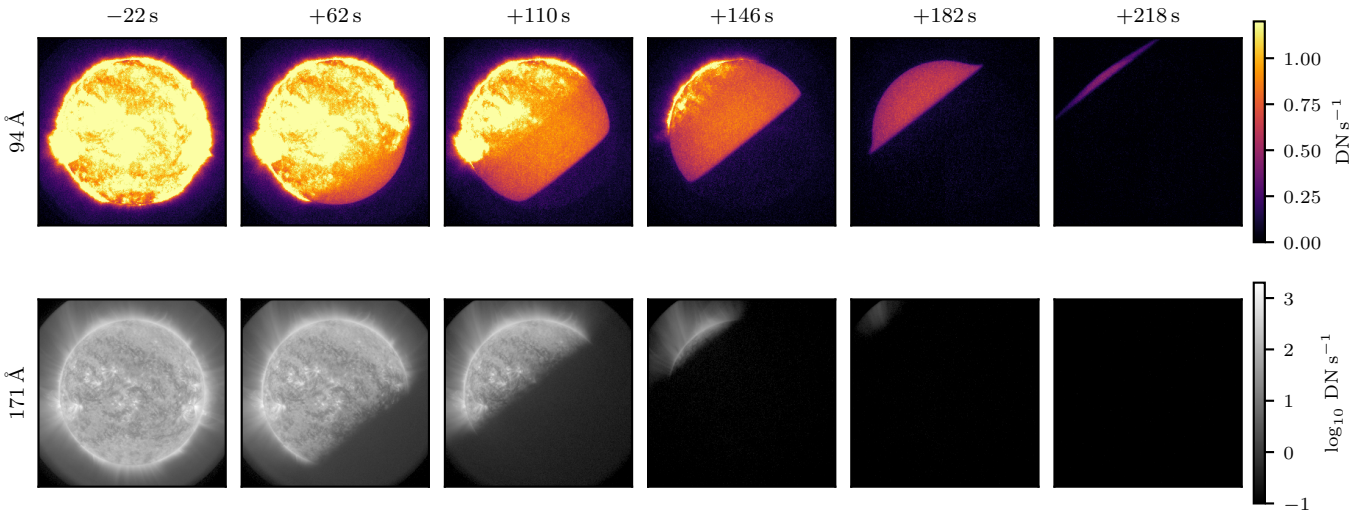


Figure 2. The ingress of the eclipse of 2026 July 15 in the 94 Å (top) and 171 Å (bottom) channels, at the times marked after the 171 Å disk mean began to fall. Each frame is level 1, averaged over 16 × 16 pixels, less the median of the umbra just after the ingress, and divided by its exposure time; 94 Å is multiplied by the flat field at disk center, so that it is in the units of the detector there. The atmosphere absorbs the EUV from the lower right, and the corona of both channels goes with it, but in 94 Å a smooth, limb-darkened disk is left behind. It sets 81 s after the EUV edge has passed, flattened by refraction, as the visible Sun does.

Henke et al. (1993), as the calibration expects. From 0.40 μm to 1.1 μm, beyond which the silicon of the detector is transparent, it is at most 0.011 with the optical constants of M. R. Querry (1987), 0.012 with those of D. W. Lynch et al. (1975), and 0.075 with those of the film of K. J. Palm et al. (2018), the least absorbing of them. So the leak is visible light that crosses the metal of the thin filter, and that the extra zirconium of the spare all but stops. Redo this comparison if the AIA team can take frames through the spare outside an anomaly of the instrument, each followed by a frame through the thin filter of 304 Å: it rests on a single frame, taken the day after a reset.

- **The entrance filters.** The zirconium entrance filters of the channel pass enough visible light to image the photosphere, and AIA monitors them with images taken without a focal-plane filter (P. Boerner et al. 2012). Without one, the light of both halves of the aperture of the 94 and 304 Å telescope reaches the detector: such an image records the visible light that reaches the focal-plane filter. The images show the photosphere, darkened toward the limb and with its sunspots, and also arcs and streaks of light that are not on the Sun and change over the years (Figure 6, top). One a month since the first, in 2011 December, they brightened by a factor of 67 in the ring at 0.85–0.9 $R_\odot$, measured by the median of its bins, which stays in the linear range of the camera (Figure 6, bottom left). In every eclipse since 2019, once the leak in that ring was 3 times its mean before 2016, it has been 3.1×10^{-7} of the brightness of these images there, with a scatter of 4.2%, while it grew by a factor of 4.6 (Figure 6, bottom right). Before, the ratio was higher, 3.8 times as high in 2012, when the leak was mostly the faint glow seen before the onset. So the leak grows because the entrance filters pass more and more visible light, of which the focal-plane filter passes a constant part. The images look more limb-darkened than the model of the leak; compare the radial profiles. The arcs and streaks cross the ring; show that the ratio does not depend on the statistic of the ring, and find where they come from.
- **The other channels.** The 304 Å channel shares the telescope, the CCD, and the visible light that reaches the focal-plane filter, but has no leak, so the leak passes the zirconium focal-plane filter of 94 Å and not the aluminum one of 304 Å. What the noise says, and cannot say, about the 131, 171, 193, 211, and 335 Å channels.

5. CONSEQUENCES

Outline.

- **Differential emission measures.** The DEM of each bin of the six coronal channels at noon of each eclipse day, inverted by the method of J. Plowman & A. Caspi (2020) with the response of SolarSoft, corrected for degradation and for the lines missing from 94 Å (P. F. Boerner et al. 2014), as observed and with the leak measured in the eclipse of the same day subtracted from 94 Å (Figure 7). On 2026 January 19, 71% of the emission measure of the coronal holes and 37% of the quiet Sun is above 3.2 MK as observed, and 14% and 3.9% with the leak subtracted. As observed, both grow with the leak, from at most 21% and 9.4% before 2016; with the leak subtracted, neither is more than 14% and 7.2% on any day. The quiet Sun is chosen by its brightness in 193 Å, and near solar maximum it is brighter in 94 Å too, so that the leak is a smaller part of it.

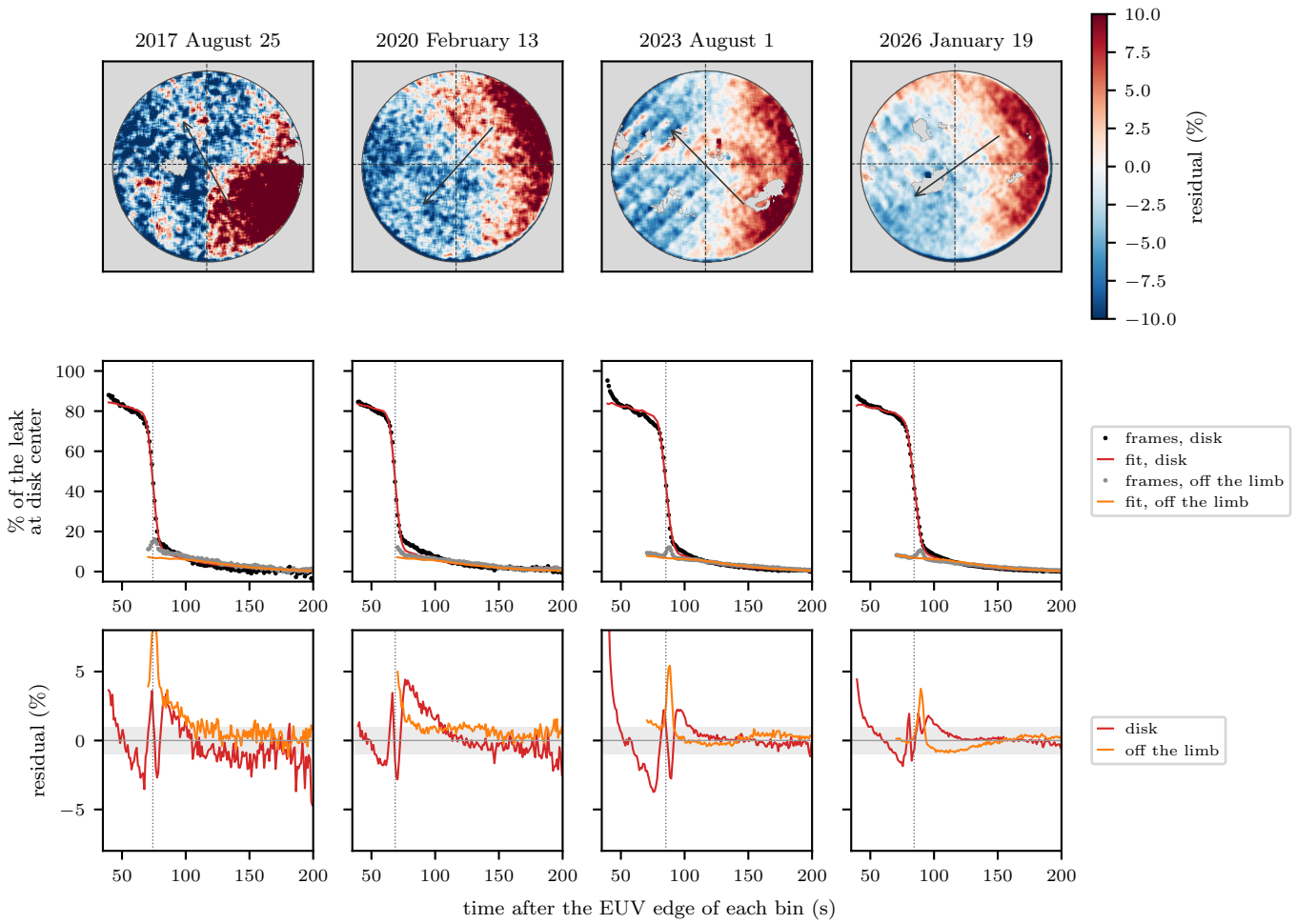


Figure 3. The fit to the ingress of four eclipses, in percent of the leak at disk center. Top: the mean residual of each bin over its plateau, from when its EUV is gone until 10 s before the visible disk sets there, averaged over squares of 5×5 bins; gray where no frame is fit, dashed at the edges of the quadrants of the detector, and the arrow the direction the EUV edge sweeps the disk. Middle: the frames (points) and the fit (lines) against the time after the EUV edge of each bin, the mean in each second of the bins on the disk (within $0.95\,R_\odot$; black and red) and off the limb (1.05 to $1.4\,R_\odot$; gray and orange). Bottom: their difference, on the disk (red) and off the limb (orange); the gray band is ±1%. The dotted line is when the visible disk is half set.

- **Fe XVIII.** In quiet regions the leak survives the subtraction of the cool part of the channel.
- **Calibration.** In full-disk comparisons a growing leak reads as rising sensitivity. The correction tables now hold the sensitivity of the channel constant. With the leak subtracted, 94 Å in the quiet Sun is 83% of what its DEM predicts in 2010 and 48% in 2026, falling between them though not steadily: the sensitivity of the channel to the EUV may have fallen while the leak hid it from full-disk comparisons. Check against the degradation of the other channels and against SUVI.
- **Published results.** Which of the results of Section 1 the leak accounts for, and how much.

6. A CORRECTION

Outline.

- **Model.** A limb-darkened disk seen through the PSF of the leak, scaled by an amplitude interpolated between eclipses and divided by the flat field at disk center: flat fielded, the leak keeps the shape the eclipses measure.
- **Uncertainty.** From the scatter about the exponential and between the methods.
- **Keeping it current.** Two eclipse seasons a year measure the leak as it grows. Decide where the correction lives: `aiapy`, `sdo`, or this package.

7. CONCLUSIONS

Outline.

- **The leak.** The 94 Å channel records visible light from the photosphere: a limb-darkened image of the disk with a broad halo, which adds almost no photon noise.

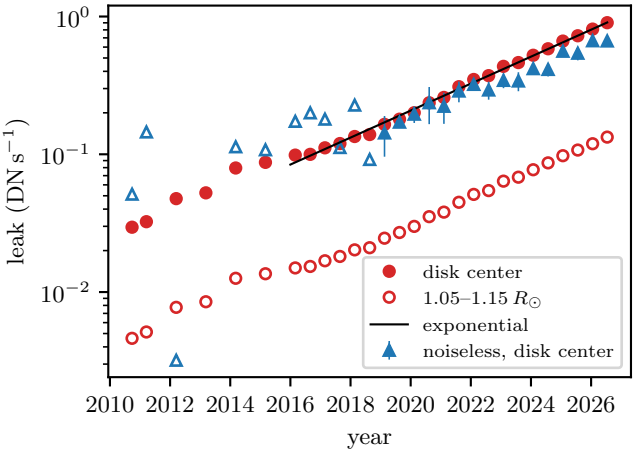


Figure 4. The leak in the 28 eclipse ingresses, flat fielded and in the units of the detector at disk center, at disk center (filled circles) and just off the limb (open circles), each with the whole disk up, and the noiseless signal near disk center in a run at noon of each day (triangles), with its spread among the quadrants of the detector; it is open where the leak is less than three times the typical spread, 0.1 DN, and the noise does not measure it. The line is an exponential fit to the eclipses since 2016, which doubles every 3.1 yr.

At disk center it grew from 0.030 DN s^{-1} in 2010 to 0.90 DN s^{-1} in 2026, doubling every 3.1 yr since 2016, and it is now 66% of the quiet-Sun signal in the channel.

- **Two measurements.** The noise of the channel and the eclipses find it independently. Since 2019 the noiseless signal is 71% to 100% of the leak in the eclipses.
- **Its path.** The zirconium entrance filters have always passed some visible light, and they pass more and more: the images taken without a focal-plane filter brightened by a factor of 67 since 2011 December, and since 2019 the leak has been a constant 3.1×10^{-7} of them. The thin zirconium focal-plane filter passes a fixed part of that light through its metal: the thick one, with 1000 Å more zirconium, stops the leak, and 304 Å, behind an aluminum focal-plane filter, has none.
- **Consequences.** Left in the data, the leak appears in DEMs as hot plasma outside active regions: on 2026 January 19, 71% of the emission measure of coronal holes and 37% of the quiet Sun is above 3.2 MK, and 14% and 3.9% with the leak subtracted. Results on hot plasma in the quiet Sun and coronal holes since 2016 should be revisited. With the leak subtracted, 94 Å in the quiet Sun falls from 83% to 48% of what its DEM predicts, so the channel may have degraded more than its calibration says.
- **What to do.** Subtract the model of Section 6, kept current by the eclipse seasons and, between them, by the images taken without a focal-plane filter. Observing through the thick focal-plane filter would stop the leak and keep 0.67 of the EUV, as long as 304 Å stays on its thin filter, since the band comes with a move between the two thick filters.
- **For the AIA team.** Why the entrance filters pass more and more visible light, what adds the band to the frames of the routine checks of the thick focal-plane filter, and where the arcs and streaks in the images taken without a focal-plane filter come from.

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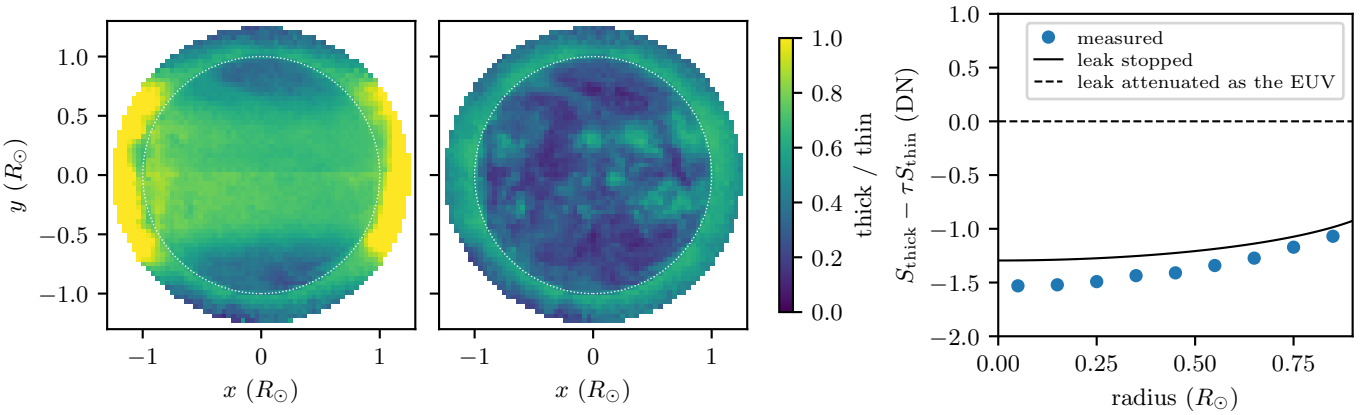


Figure 5. The two focal-plane filters of 94 Å on 2026 March 7: the last two frames through the thick filter, as the channel returned to the thin one, and the first 15 frames through the thin filter, which began 24 s after the last. Left: the first of the two frames through the thick filter, read out as the filter wheel moved to the thick filter of 304 Å, over the frames through the thin one, in each region of 64×64 pixels; it carries a band across the middle rows of the detector. Middle: the same for the last frame, read out as the wheel moved to the thin filter of 304 Å. Right: the signal of the last frame through the thick filter less $\tau = 0.67$ times that through the thin one, the median of each ring of $0.1\,R_\odot$ (points), and what it would be if the thick filter stopped the leak (solid) or attenuated it as much as the EUV (dashed).

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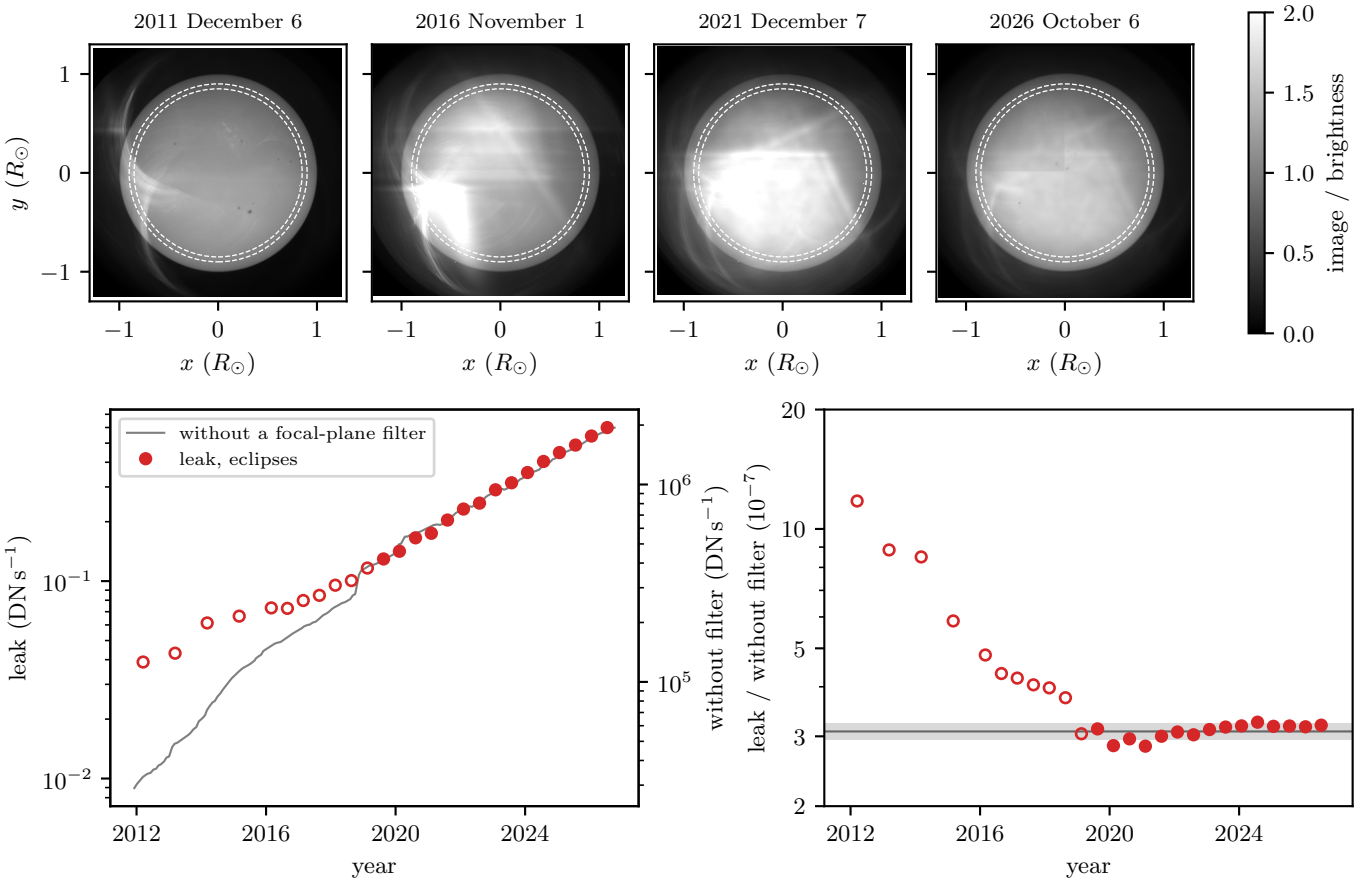


Figure 6. The images of the 94 and 304 Å telescope taken without a focal-plane filter, one a month since 2011 December, against the leak. Top: four of the images, from the first to the last, in bins of 16 × 16 pixels, each divided by its brightness, the median of its bins in the ring at 0.85–0.9 $R_\odot$ (dashed). Bottom left: the leak in each eclipse (red), through the point spread function fit to the eclipses, and the brightness of the images (gray, right axis), scaled by the mean ratio of the two, both in the ring and in the units of the detector. Bottom right: the ratio of the leak to the brightness of the images at the time of each eclipse, with its mean and scatter (gray) over the eclipses since the leak in the ring was 3 times its mean before 2016 (filled).

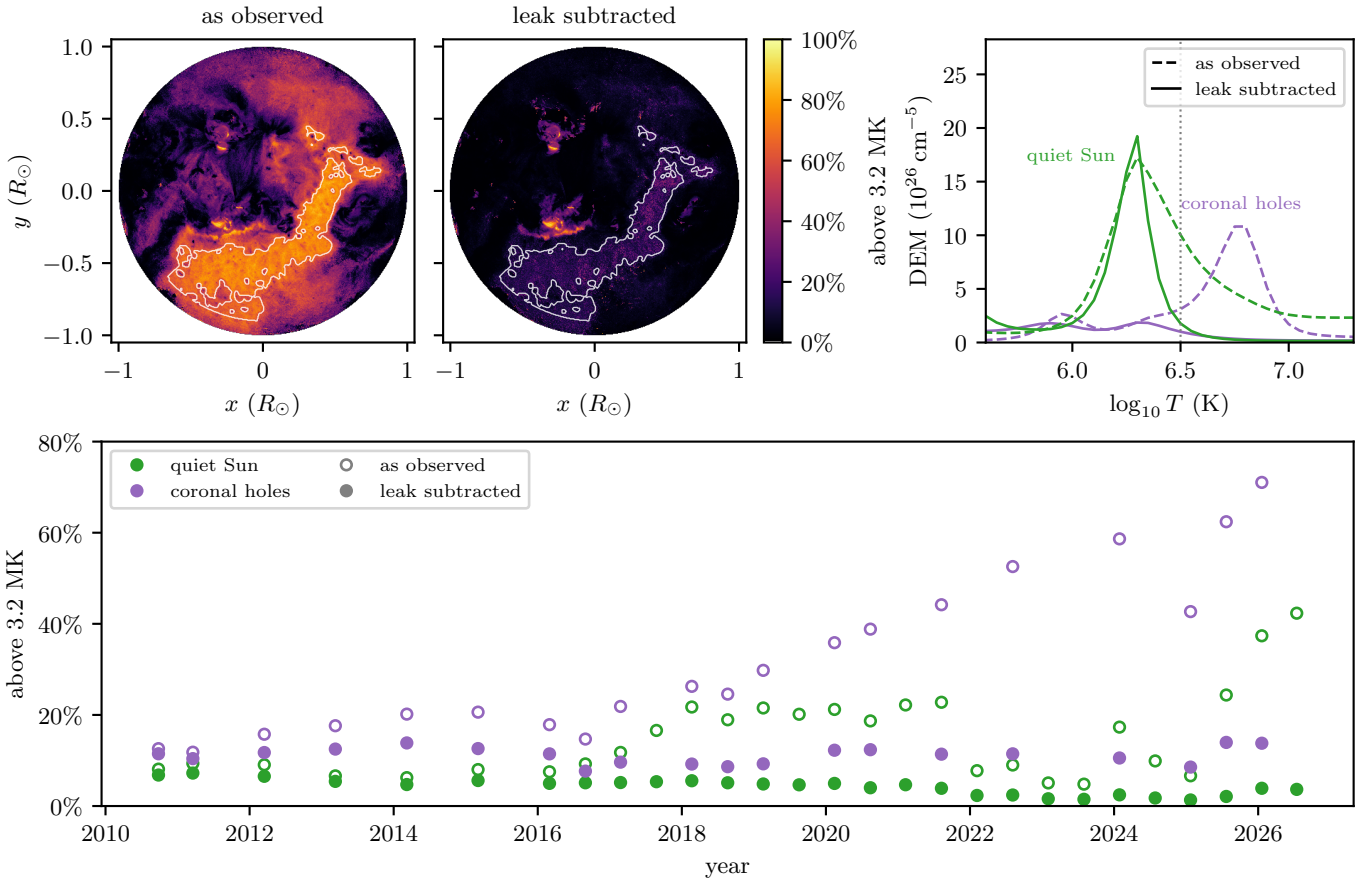


Figure 7. The DEM of each bin, 4.8″ on a side, inverted from the six coronal channels at noon of each eclipse day, as observed and with the leak measured in the eclipse of the same day subtracted from 94 Å. Top left and middle: the fraction of the emission measure above 3.2 MK on 2026 January 19, as observed and with the leak subtracted, on the disk where the inversion succeeded; the coronal holes are outlined in white. Top right: the median DEM of the coronal holes and of the quiet Sun that day, as observed (dashed) and with the leak subtracted (solid); the dotted line is 3.2 MK. Bottom: the median fraction above 3.2 MK in the coronal holes and the quiet Sun of every eclipse day, as observed (open) and with the leak subtracted (filled), with the coronal holes shown where they cover at least 500 bins.